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Final Project Write Up

BTC Prediction Market Fraud Detection Using Statistical Anomaly Detection and LLM Assisted Reasoning

1. Introduction

Prediction markets allow various users to trade contracts as well as dollar amounts on real world outcomes providing probabilities for events ranging from sports and weather to crypto-currency price movements. Kalshi which is my platform of choice is one of these new exchanges that recently came to view and quickly became a known platform for trading for contracts listed above including the focal point of this project, Bitcoin and whether it will increase or fall short of a certain price range given some date. However, given the volume being traded and how easy it is to trade said contracts this market especially has become vulnerable to manipulation tactics which I tried to explore and document. The three main points I focused on during this project which are also the main forms of manipulation include wash trading, coordinated trade clustering and event driven changes. For starters wash trading involves someone and something trading against itself to artificially inflate volume and create a misleading idea of market activity. Coordinated trade clustering happens when an agent or group of individuals make rapid but biased trades to shift a contract's implied probability. Lastly, event driven changes happen when large positions are placed based on non-public information about future price movement and events. These tactics are used consistently, and many are not spotted which further influences these users to partake in these tactics. Though efforts have been made and fall into two categories. First one being a rule-based system which flags trades that exceed fixed thresholds. These thresholds lack the ability to reason about context. Some machine learning systems can learn these patterns but require large, labeled datasets that don't exist for these types of markets. These approaches again, don't suit this type of market considering all characteristics. This project aims to explore the methods mentioned above and test them to see how they perform along with an ai interpretation of said data and how that can help influence decisions for analyst and security experts.

2. Background/Previous work

Regarding fraud in financial markets, detection systems have been researched for decades with methods going beyond threshold-based rules to more

complex machine learning systems. Early methods consisted of statistical process control methods, applying z score analysis and moving averages to identifying weird movements. These methods are still around because of how easy it is to interpret and low cost. Though times have changed and the focus has moved onto deep learning approaches. One example mentions how deep learning consists of multiple hidden layers which include interconnected neurons organized in layers. This is made up of the input, hidden and output layers. Then the info flows through the network via weighted connections and a bias term is added to the weighted inputs and the sum is passed through an activation function. This is surface level but gives a good representation as to how the tech for these types of markets expanded and evolved and how my approach differs. I chose approaches most commonly seen in crypto markets specifically as it would be an interesting insight since crypto works much differently than stocks. My idea was to build on this and add a better way of reasoning through the use of LLMs. These models can interpret the data, generate explanations and condense it into a digestible format for easy interpretability. Some challenges that these models face is the sheer amount of data they need to ingest which can be overwhelming.

3. Methodology

Starting from the very first process this detection pipeline consists of four stages first one being data ingestion from Kalshi's API, moving onto the anomaly detection across four categories, then the evidence brief is made which compresses the information into a structured summary and lastly into the LLM assisted analysis generated using Ollama but can also be used with other paid for alternatives for higher accuracy. For easier start I included a synthetic dataset made to test the anomalies quickly, locally and without any issues. This was made up of 178 generated trades simulating a BTC market over 10 hours. It had two phases, one with normal looking trades and the other which had three injected fraud phases. This included a 25-trade volume spike, an 18-trade wash trading sequence and a 15- trade cluster with 100% directional bias. The live data consisted of 5560 real trades taken from Kalshi's API across five BTC markets spanning January 12th to February 27th of this year. Each trade was made up of a trade ID, market ticker, contract quantity, YES/NO price, taker side, and timestamp. Now I'd like to take the time to talk about the specific algorithms used for detecting such anomalies. These systems use four deterministic statistical detectors which means there is no training phase or learning curves. Each of these targets a specific manipulation category. Starting with the volume spike detector. For this detector trades are grouped into non overlapping five-minute windows. The total

contract volume per window is z scored against the distribution of all windows and a window is flagged if the z is greater than or equal to 2.0 and is marked as medium severity. If the z is greater than or equal to 3.0 then it is marked as high severity. This detects weird surges in trading activity that deviate from the baseline movement. Next is the price swing detector. For each consecutive pair of trades sorted chronologically the percentage change in yes price is computed. For swings greater than 15 percent are flagged as medium, if greater than 30 then it's marked as high. A known limitation is that binary contracts trading with a large change produce inflated percentage changes from small movements which is something to look out for. The wash trading detector works by taking two sub methods and combines them to work together. First the trades are grouped by exact pairs which is determined by price and quantity. Then if three or more identical combinations appear within a 600 second window the pattern is flagged as a repeated combination. Second the detector scans for runs of five or more consecutive trades with alternating taker side suggesting one entity is on both sides of the trade. Following that we have trade clustering which is a sliding window of 300 seconds and identifies windows with five or more trades. For each cluster the directional bias is computed as the percent of yes or no and is divided by the total. A high cluster size combined with strong directional bias might suggest that a coordinated buy/sell occurred. If the cluster contains more than 10 trades, then it's flagged as high severity and medium otherwise. The last method is a composite risk score which is calculated by multiplying high flags by 25 plus medium flags times 10 and is capped at 100. This provides a summary metric for overall market health. Now, after the data runs through the detectors it's flagged by severity within each category and the top five most significant per category are combined into a structured block. This includes some market context, statistics, and per flag details that further explain what's happening. These blocks are useful for the LLM because it's a compressed version so it's not ingesting the full thing while preserving what's most important. In this scenario though anyone can modify it to use whatever LLM they'd like, I went with the free Ollama version that I can run the data through. This LLM classifies each flag as actual fraud, suspicious or likely nothing to worry about and then It will return a final verdict with the risk level, and further recommendations.

4. Experiments

Starting with the synthetic dataset, this was used to validate that each detector works and identifies the fraud correctly through various patterns. As mentioned above we know exactly what the fraud was in this case because it's only a test. So, in summary total trades analyzed was 178, total flags raised was 34, amount of High severity flags was 5, Medium severity was 29 and the score was 100/100. This also

means detection rate was 100% and the False positive rate was 20%. All patterns were successfully detected. I'd also like to include that the volume spike at hour 4:30 produced a z-score of 7.86 which indicates that the flagged window contained around 28 times the average trading volume. This window was also flagged by the cluster detector. The wash-trading injection produced two high severity repeated combo flags which includes $0.70 × 10.00 pair appeared 12 times in 90 seconds and the $0.68 × 5.00 pair appeared 6 times in 60 seconds. The coordinated cluster injection was detected as 15 trades in 86 seconds with 100% yes directional bias. Seven medium flags were raised on the organic baseline phases producing a false positive rate of 20.6%. These occurred from random clustering which are expected statistical artifacts at these threshold levels. Moving onto the live Kalshi data after the full pipeline was applied to the 5560 trades these were the results: Volume spike containing 10 total flags all being high severity with a z-score of 13.92, which was the most significant find. Next up is the price swings which contained a total of 751 flags of which 279 were flagged as high severity and 472 as medium severity. A 900% swing was the most significant finding for this detector. For the wash trading detector there was 207 total flags with 153 being high severity and 54 being medium severity with the most significant movement being 37 repeats in 4 min. Lastly, the trade cluster detector picked up 263 total flags with 172 being of high severity and 91 being medium severity. The most extreme movement was 162 trades in 5 minutes. However, many of these flags were from small changes near extreme prices which produce inflated percentages. With this in mind I want to focus on the LLM aspect of this project. The evidence brief from the live data analysis was submitted to a locally hosted Ollama model. The LLM correctly identified the volume spike and coordinated clusters as the most likely indicators of genuine manipulation classifying them as likely manipulative activity. It appropriately labeled the price swings as suspicious but inconclusive. It assessed the wash trading indicators as suspicious noting that further investigation would be required for confirmation. The LLM's final verdict was high risk with 85% confidence recommending a review of exchange records collaboration with Kalshi staff and targeted investigation of the volume spike and clustering events.

5. Discussion

The most significant finding I thought was most valuable is the cross-pattern convergence. For example, in the synthetic data the injected volume spike was independently flagged by three of four detectors on the same 5-minute window. In the live data February 6, 2026, all four detectors converge on a two hour trading

session. The probability of four independent statistical tests triggering on overlapping windows by chance is very low making multi-detector convergence the strongest fraud signals the system can produce. This does have limitations, nonetheless. Without specific account level data, the wash trading detectors cannot confirm whether repeated combos originated from the same entity. The price swing detector is noisy for low priced contracts producing a high volume of flags that dilute the overall signal to noise ratio. The composite risk score saturates at 100 reducing its ability to differentiate between moderately and severely anomalous markets. Additionally, the system cannot distinguish between genuine manipulation and legitimate reactions to external events. The LLM layer attempts to address this gap by reasoning about context but without integrated news feeds its ability is limited to say the least. The LLM integration using Ollama demonstrated that a locally hosted model can produce useful analysis of statistical findings at zero marginal cost. However, the quality of reasoning was noticeably below that of other larger models. Specifically, the local model missed the cross pattern convergence between the volume spike and trade clustering on the same time window and it classified wash trading as merely suspicious when the 37 repeat pattern is a clear indicator of automated self dealing. This suggests that local LLMs are viable for the pipeline, model selection and prompt engineering have an impact on analysis quality.

6. Future Work

As for the future of this project and the topic in general I plan to keep up and continue tweaking it for the better. Financial limitations are up there as of right now but that aside this can be improved. As for the industry and markets the future is promising because tweaks can be made further boosting accuracy and model reasoning. Thinking in terms of my project some things I would considering doing to further improve it consist of implementing adaptive thresholds for the price swing detector which would reduce false positives on low priced contracts. Instead of a 15% threshold a dynamic function that scales with contract price would ensure that only genuinely anomalous swings are flagged. In addition, integrating a news API would allow the system to cross reference flagged time windows with known BTC events enabling the LLM to distinguish between event driven trading and manipulation. Another improvement which could significantly improve results would be to improve the LLM itself. Looking into a larger model could improve reasoning quality both local and not. Though models that are not ran locally could be much more expensive.

7. Conclusion

In conclusion, through this paper I tried my best to present a fraud detection system for Bitcoin markets that uses statistical anomaly detectors paired with LLM assisted reasoning. The pipeline consisting of four detectors showed interesting results using both synthetic and live Kalshi data. The evidence brief layer that takes the data and compresses it into a digestible format for the LLM demonstrated that LLM reasoning can be effective to an extent. This project aimed at quantifying suspicion and identifying anomalies consistent with known patterns and gives analysts an insight and a reason to investigate.

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